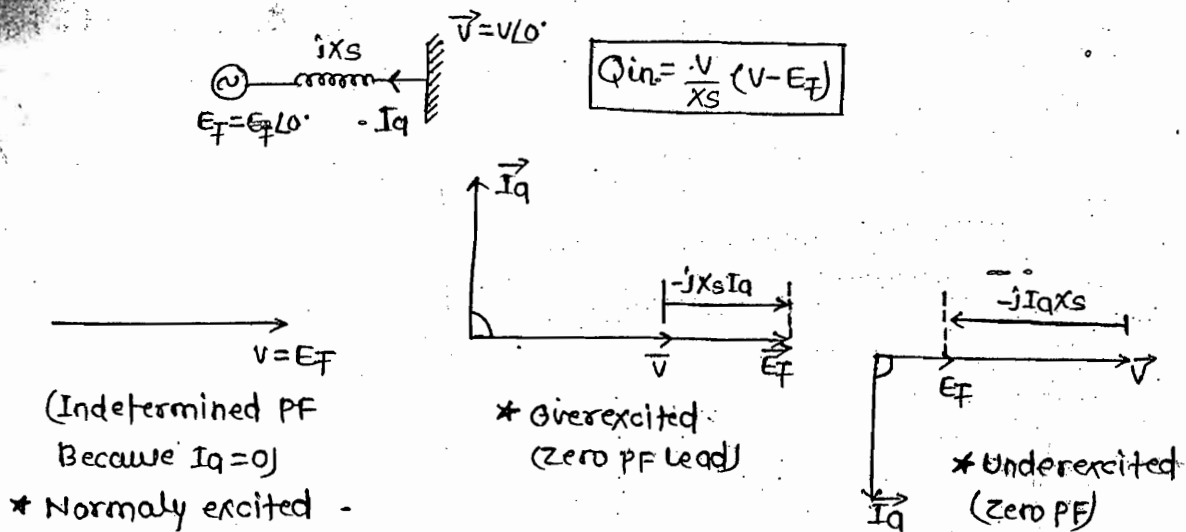


* Synchronous Condenser $\rightarrow$ (synchronous phase advancing, syn. capacitor)



- * Underexcited - absorbing lagging VARs.
- * syn. compensator (actual name)
- * syn. phase modifier

Note:- (1) If big m/c is operating without prime mover & shaft $\rightarrow$ syn. com.

(2) The dia. of shaft is kept small because of operating on NL.

(3) It is used 200 MVARs (upto)

(4) It gives clean Reactive powers (like w/o harmonics)

Gen^r $\rightarrow Q_{out} = \frac{V}{X_s} (E_f \cos \delta - V)$

If $E_f = 0$, $\cos \delta = 90^\circ$

$Q_{out} = \frac{-V^2}{X_s}$

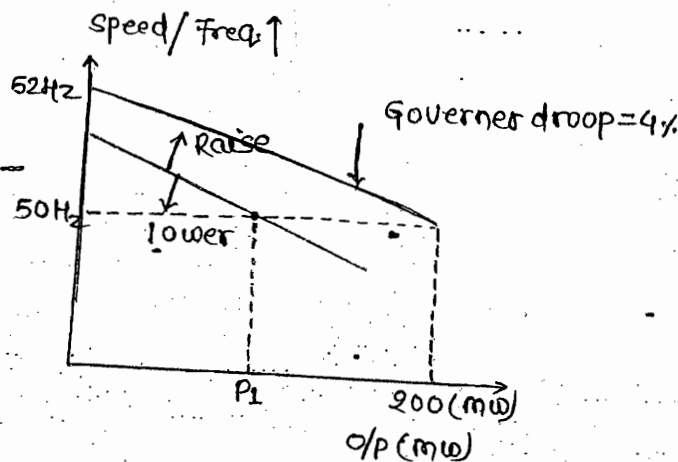
$Q_{out} = -V^2 \times SCR$

(line charging capacity)

- * Turbogener are situated at load center so req. less X_n line & the SCR is low but hydrogener is situated at different area so req. more X_n line & SCR is high for those.

* Parallel operation of gen^r →

* The ele. aspects of parallel operation of syn. gen^r would be handled by Millman's theorem exactly similar as in parallel operation of TF with unequal voltage ratio.



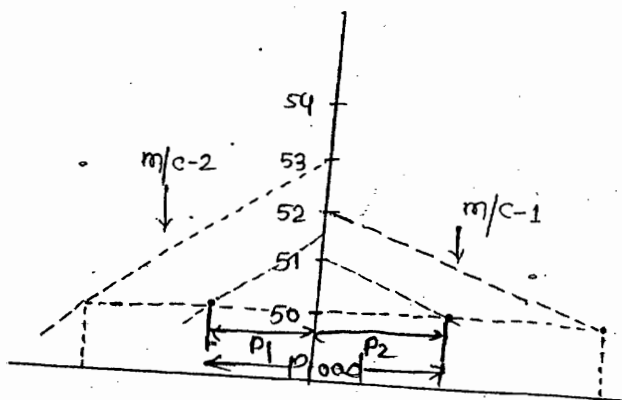
* Governor Regⁿ = $\frac{52\text{Hz} - 50\text{Hz}}{52\text{Hz} - 50\text{Hz}} = \frac{2\text{Hz}}{50\text{Hz}} = 0.04 \text{ pu} / 4\%$
(OR)

Droop = $\frac{52\text{Hz} - 50\text{Hz}}{200\text{MW}} = 0.01 \text{ Hz/MW}$

when once droop is decided then it is not changed.

* Control gear also known as speeder gear/ speed changer is a device that simply shifts the governor c/s 11^{rad} to itself on being given a raise (or) lower command. (above dia.)

In pu (droop) = $\frac{1.04 - 1.00}{1.00} = 4\%$



Que → 2 identical 2000kW, 50Hz alternator operate in parallel. The governor of the 1st m/c is such that the freq. rises uniformly from 50Hz on FL to 52.5Hz on NL. The corresponding uniform freq. rise of 2nd m/c is 50Hz to 52.5Hz.

(a) If each m/c is fully loaded at rated freq. what would be load in each m/c when total load is reduced 2560kW.

(b) Cal. the max^m load at which one of the m/c would become unloaded?

Solⁿ → (a) →

m/c A

$$\frac{P_A}{2000} = \frac{52 - F}{52 - 50} \text{ --- (A)}$$

m/c B

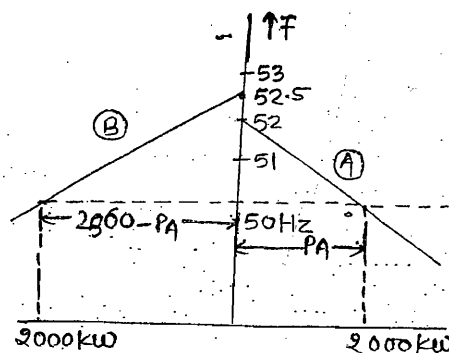
$$\frac{2560 - P_A}{2000} = \frac{52.5 - F}{52.5 - 50} \text{ --- (B)}$$

Solving eqn (A) & (B)

$$P_A = 1200 \text{ kW}, P_B = 2560 - 1200$$

$$= 1360 \text{ kW}$$

$$F = 50.8 \text{ Hz}$$



(b) →

$$\frac{P_L}{2000} = \frac{52.5 - 52}{52.5 - 50}$$

$$P_L = 4000 \text{ kW}$$

Que → 2 identical 2000kW alternators operate in parallel. The governor of the 1st m/c is such that the freq. drop uniformly from 50Hz on NL to 48Hz on FL. The corresponding uniform freq. drop of 2nd m/c is 50Hz to 47.5Hz.

(a) How will the 2-m/c share the load of 2880kW.

(b) What is the max^m load that can be delivered w/o overloading either m/c.

Solⁿ →

q) →

$$\frac{m/c A}{2000} \frac{P_A}{2000} = \frac{50-f}{50-48} \text{ --- (A)}$$

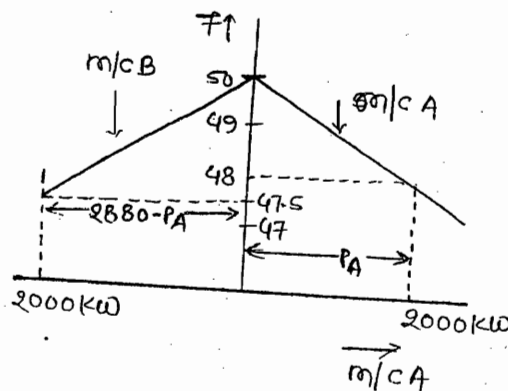
$$\frac{m/c B}{2000} \frac{2880-P_A}{2000} = \frac{50-f}{50-47.5} \text{ --- (B)}$$

Solving eqn (A) & (B)

$$P_A = 1600 \text{ kW}, f = 48.4 \text{ Hz}$$

$$\text{Then, } P_B = 2880 - 1600$$

$$= 1280 \text{ kW}$$



(b) →

$$\frac{P_B}{2000} = \frac{50-48}{50-47.5}$$

$$P_B = \frac{2000 \times 2}{2.5} = 1600 \text{ kW}$$

$$P_{\text{load (max)}} = 2000 + 1600$$

$$= 3600 \text{ kW}$$

$$P_{\text{load (max)}} = 3600 \text{ kW}$$

* Synchronization →

Condition to be satisfied →

- (1) Equal freq.
- (2) Equal voltage
- (3) Same polarity in 1- ϕ gen^r & same phase seq. in 3- ϕ gen^r.
- (4) Zero phase difference at instant of synchronization.
- (5) Same waveforms.

Synchronization by synchroscope →

- (1) match the freq. of the incoming m/c with that of the running m/c. It is recommended that the incoming freq. be kept slightly higher.

* (2) match the incoming vol. with the running vol. It is again recommended that the incoming vol. be kept slightly higher

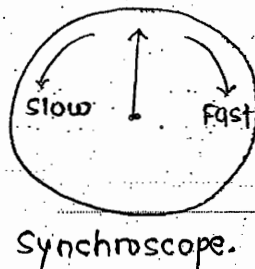
* (3) Switch ON the synchroscope & also the check synchronization relay if such relay is available. Switch off anti motoring protection if provided.
(Reverse power protection)

* (4) Fine control the incoming freq. such that the synchroscope pointer moves very slowly in the fast dirn. when the pointer reaches 11 o'clock position, give a command to the CB switch to close the breaker.

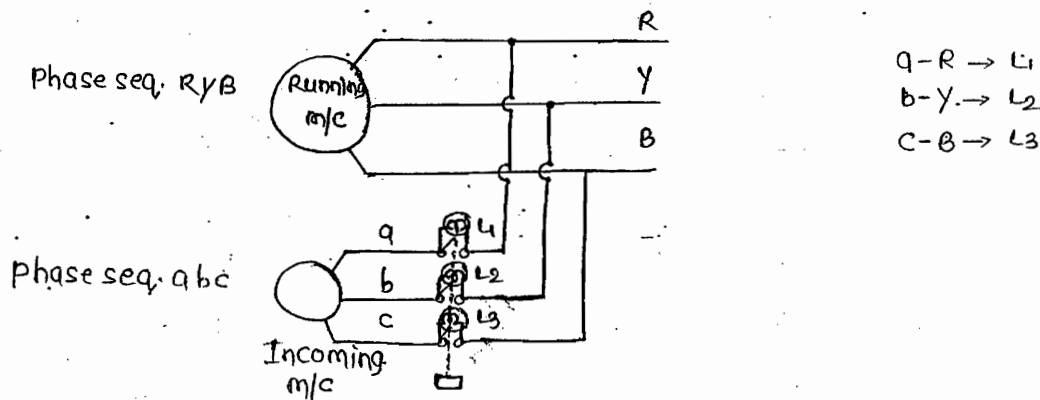
It is expected that the breaker contacts would actually close at 12 o'clock position, a position that represents zero phase diff.

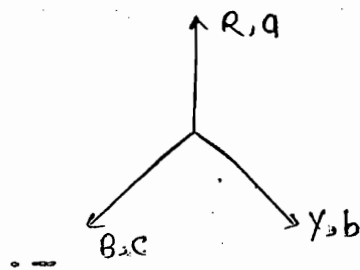
* (5) Switch-off the synchroscope & also check synchronization relay. Take min^m recommended load on the unit & adjust excitation for desired PF of operation.

* (6) After taking sufficient load on the unit switch on the anti motoring protection switch.



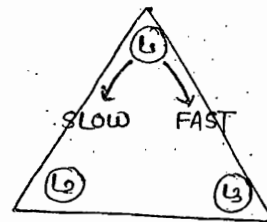
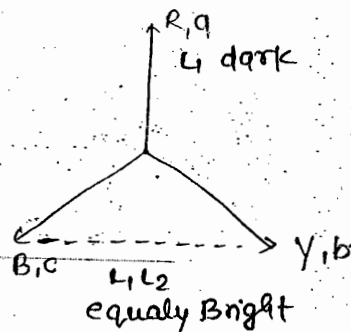
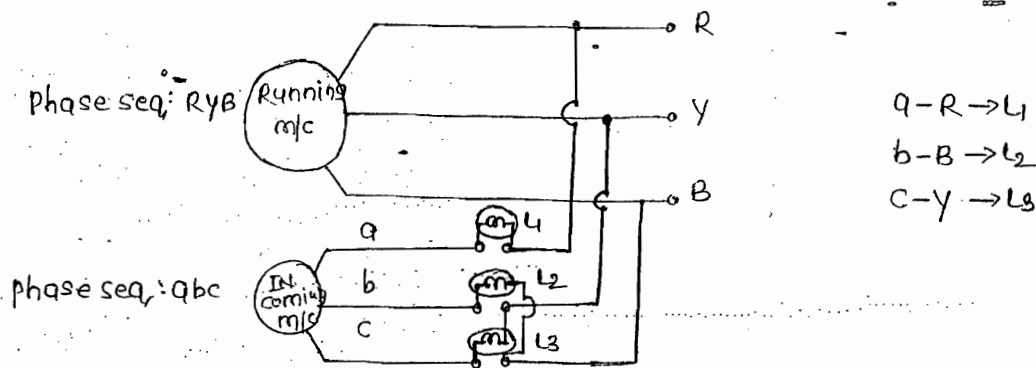
* Dark lamp method → (Kothari/Ashfaq)





L_1, L_2, L_3 dark then synchronise.

* Rotating Lamp method $\rightarrow$ (Dark & bright method) (Siemens & HALSKE method)



Sequence $\rightarrow$

- $L_1 - L_3 - L_2$; Incoming m/c is Fast.
- $L_1 - L_2 - L_3$; Incoming m/c is slow.

5.10/600

Que. $\rightarrow$ A syn. m/c is synchronised with an ∞ bus at rated vol. Now the steam i/p to prime mover is increased. The syn. m/c starts operating at rated kVA. The m/c has syn. imp. $z_s = 0.02 + j0.8$ pu. Determine the operating pf of alternator & its load angle.

Analytical approach →

Solⁿ →

$$Z_s = 0.02 + j0.8 \text{ pu}, \text{ PF} = ?, \delta = ?$$

$$= 0.80025 \angle 88.57^\circ \quad \cos \phi = ?, \delta = ?$$

$$E_f = V + I_a R_a ; \quad \cos \phi = \frac{\text{kVA}}{VI}$$

$$E_f = 1 \angle \delta, V = 1 \angle 0^\circ, I_a = 1 \angle \phi$$

~~22~~

$$1 \angle \delta = 1 \angle 0^\circ + 1 \angle \phi (0.80025 \angle 88.57^\circ) \dots$$

$$1 \angle \delta = 1 + 0.80025 \angle (88.57^\circ + \phi)$$

$$0.80025 \angle (88.57^\circ + \phi) = 1 \angle \delta - 1 \quad \text{--- (i)}$$

$$0.80025 \angle (88.57^\circ + \phi) = (\cos \delta - 1) + j \sin \delta$$

squaring & equate eqn

$$(0.80025)^2 = 1 - 2 \cos \delta + 1$$

$$\boxed{\delta = 44.17^\circ}$$

Putting in eqn (i)

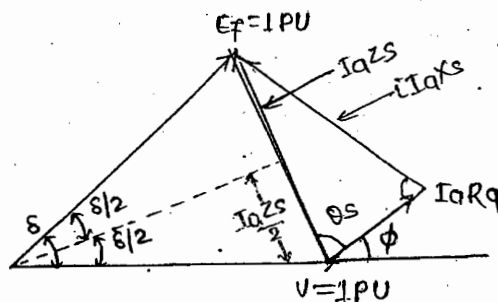
$$(88.57^\circ + \phi) = \frac{\angle [1 \angle 44.17^\circ - 1]}{0.80025} = 113.59^\circ$$

$$\phi = 25.02^\circ$$

$$\text{PF} = \cos(25.02^\circ) \text{ leading}$$

$$\boxed{\text{PF} = 0.9062} \text{ leading}$$

Graphical Approach →



$$\theta_s + \phi = 90^\circ + \frac{\delta}{2}$$

$$\phi = (90 - \theta_s) + \frac{\delta}{2} \text{ (leading)}$$

$$\sin \frac{\delta}{2} = \frac{I_a Z_s}{V}$$

$$\frac{\delta}{2} = \sin^{-1} \left[\frac{1.0 \times 0.80025}{1 \times 1.0} \right]$$

$$= 23.586^\circ$$

$$\delta = 47.17^\circ$$

$$\phi = (90^\circ - 88.57^\circ) + 23.586^\circ$$

$$\boxed{\phi = 25.02^\circ}$$